



The Wow! signal

Scientists now believe that cometary activity gave rise to an apparently artificial signal 40 years ago. <https://digit.in/aug20-07>



Strange X-Rays

Mysterious X-Rays detected by instruments could theoretically be formed by the decay of dark matter. <https://digit.in/aug20-08>

times over, and even at this moment, there are a few open questions of exactly what the sources behind some mysterious signals from space are.

REPEATING FRBS

Fast radio bursts are intense outpourings of radio waves where an astronomical object releases the same amount of energy in millisecond bursts as the Sun does in 80 years. In these few milliseconds, an FRB can outshine an entire galaxy.

FRBs are mysterious enough to begin with, being random and unpredictable, their discovery being a matter of having a telescope lucky enough to be pointed in their direction. Most FRBs are observed only once. Since 2012, however, there have been a few detections of repeating FRBs leaving Scientists struggling to explain what could cause such massive outpourings of energies to repeat, even theoretically. In 2019, researchers from the Canadian Hydrogen Intensity Mapping Experiment (CHIME) telescope reported the discovery of eight repeating FRBs. These repeating FRBs are actually beneficial to scientists as they can study them over time and compare them. Their periodicity and strength can be observed to find out trends that may one day lead to an explanation.

One of the most well-studied repeating sources is FRB 180916.J0158+65. During a four day window, there are one or two bursts observed, followed by sixteen days of radio silence, after which the pattern repeated. This pattern has been consistently observed over a period of 500 days, during which time 38 radio bursts were observed from the same source.

There is no consensus on an explanation of what kind of objects cause FRBs, but there are a few theoretical explanations, all of which are centered around neutron stars, that if you remember, are objects made up mostly from neutrons, the remnants of novae. Rapidly rotating neutron stars are pulsars.

FRBs can be observed from a collision between a neutron star and a black hole, or hyperflares from neutron stars with an especially powerful magnetic field, called magnetars. Other possible explanation include blitzars, which are pulsars that can rapidly collapse into black holes, or pulsars collapsing on interaction with dark matter. Another possibility is a binary system of neutron stars, with a more massive star kneading and deforming the other star, causing the bursts every time the two objects are close together. Another possibility is a binary system where a radio source passes through a gas cloud that amplifies the same point in each

there is no consensus on the explanation, there is scientific consensus that the excess exists and is observable.

One of the possible explanations is that there is an undetected population of millisecond pulsars (MSPs) in the region. Not a single one, but multiple such stars. However, instead of a diffused source such as a gas cloud, analysis of the individual photons from the region showed that they were clumped together, indicating that they originated from point sources, which is in support of the pulsar hypothesis. However, this then opens up the question of why these MSPs, which are bright objects, not been detected yet. Another

explanation is that the supermassive black hole at the centre of the galaxy is attracting stars towards it, which in turn is forming an accumulation of MSPs in the region. The thing is, for MSPs to form, they have to be spun up like a top by being the stellar companion. Before it becomes an MSP, a star is called

a low-mass X-ray binary (LMXB). The proportion of MSPs in the galactic centre should be similar to the proportion of observed LMXBs. However, observations have shown that there are very few LMXBs in the region. Millisecond pulsars are however the more conventional of the proposed explanation, the other being that the gamma-rays are produced by millions of annihilation events where dark matter interacts with regular matter. The theory was discarded after the discovery of the point sources, but has been revived recently. Apparently, the existence of Fermi Bubbles, enormous (25,000 light years in diameter) spheres of gas and cosmic rays on both sides of the galactic disk, may make the radio waves from the annihilation events appear like point sources. At this point, scientists simply do not know. 



Credit: NASA

Fermi Bubbles on either side of the Milky Way

orbit. The data from astronomical instruments around the world on observed repeating FRBs are being analysed by AI so scientists can find out more about them.

GALACTIC CENTER GEV EXCESS

From around the centre of the Milky Way galaxy, there is an excess of gamma rays measured as compared to what should be seen based on the known objects in the region. If you make a list of all the known gamma-ray sources in the region, then the measured amount of radiation is more than what is accounted for by the known sources. This means that some as yet unseen source in the region is producing these gamma-rays. Even more bafflingly, the excess originates uniformly from the inner galaxy and does not seem to be related to any specific source. While